CARE-COM: Toward More Capability-Aware MoE Expert Consolidation

Investigating capability-geometry signals for more informed expert consolidation under aggressive compression.

Mixture-of-Experts (MoE) architectures achieve superior training efficiency, but their massive aggregate parameter footprints create severe memory bandwidth bottlenecks and deployment costs during high-concurrency inference. Existing MoE merging methods decide which experts to combine using static heuristics or unguided router clustering, risking destructive parameter interference. **CARE-COM** investigates whether **functional signals** from expert representations can guide expert consolidation and ultimately help identify risky merges without retraining.

SCOPE: OLMoE-1B-7B; WikiText-2 perplexity evaluation; MMLU results exploratory (N=100).

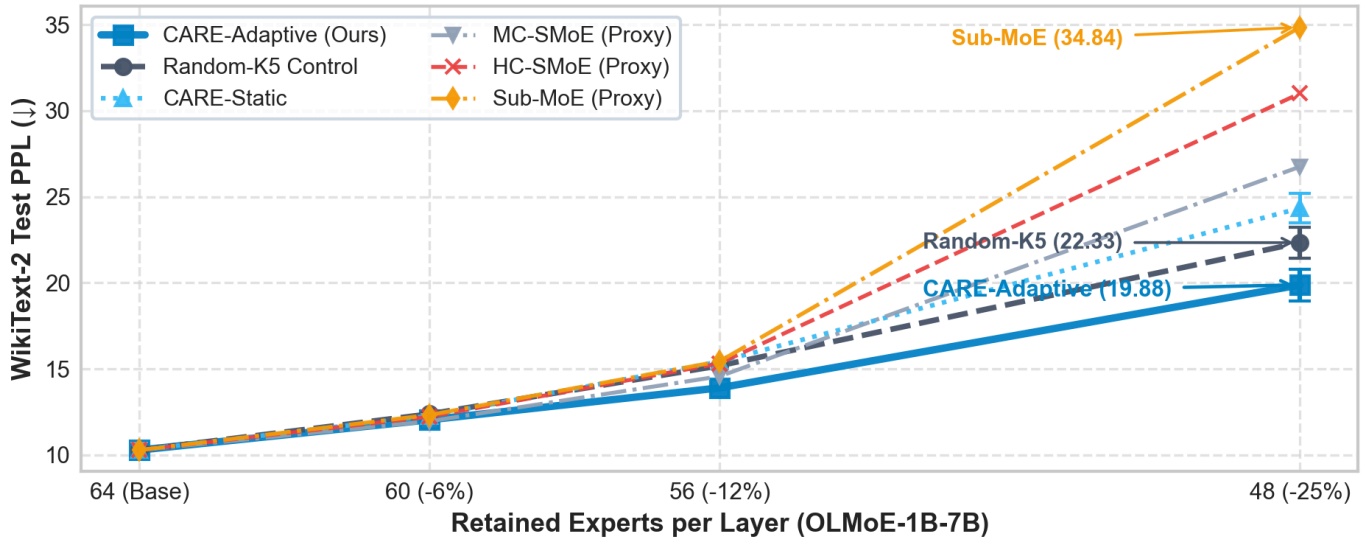
19.88 PPL
BEST OBSERVED MERGING PPL @ 48E
CARE-Adaptive (-25% budget, directional, n=3)

-2.45 PPL
MARGIN VS. RANDOM-K5 CONTROL
Directional multi-seed gain (19.88 vs 22.33 PPL)

6.9–15.0 PPL
IMPROVEMENT VS. CLUSTERING PROXIES
Vs. MC-SMoE (26.74), HC-SMoE (31.00), Sub-MoE (34.84)

THE CARE-COM PIPELINE: CAPABILITY-AWARE CANDIDATE SELECTION





OLMoE-1B-7B language modeling perplexity across Seeds 42, 43, 44 on WikiText-2 (65,536 tokens, exact token-weighted cross-entropy). External proxies represent zero-shot offline implementations. Error bars show ±1σ across seeds. Lower is better.

THE THREE SCIENTIFIC PILLARS

1. The Problem: Merging Without Damage Estimation
Existing methods merge experts based on router co-activation affinity or static weight distance, without predicting functional interference. Under aggressive consolidation (48 experts, -25% reduction), heuristic clustering baselines collapse (Sub-MoE degrades to 34.84 PPL; HC-SMoE to 31.00 PPL).

2. Offline Damage Geometry & Motivation
Offline metric analysis (Exp 4) reveals that the physical merge damage matrix has an intrinsic low-rank structure (q=4 metric MDS captures 75% of damage rank variance). This demonstrates that merge interference is geometrically structured, motivating future research on finding genuinely pre-merge observational approximations that do not require prior merge interventions.

3. Multi-Seed Compression Results
At 48 retained experts on OLMoE, CARE-Adaptive achieved the best observed multi-seed perplexity among the evaluated merging approaches: **19.88 ± 0.923 PPL** versus **22.33 ± 0.916** for Random-K5 (directional, n=3). Without dynamic loss feedback, frozen CARE-Static achieves **24.33 ± 0.871 PPL**.

MULTI-SEED COMPRESSION BENCHMARK (OLMOE @ 48E)

Method	Paradigm	PPL (μ ± σ)	Held-out KL	Δ vs Base
Uncompressed Base	Reference (64E)	**10.2804**	0.0000	Baseline
CARE-Adaptive	Dynamic Geometry	**19.88 ± 0.923**	**0.658 ± 0.038**	+9.60
Random-K5 Control	Unbiased Merge	**22.33 ± 0.916**	0.823 ± 0.043	+12.05
CARE-Static	Frozen Geometry	**24.33 ± 0.871**	0.832 ± 0.051	+14.05
MC-SMoE (Stages 1–4 Proxy)	Correlation Matching	26.74 ± 0.000	0.955 ± 0.000	+16.46
HC-SMoE (Router Proxy)	Hierarchical Clustering	31.00 ± 0.000	1.082 ± 0.000	+20.72
Sub-MoE (Joint SVD Proxy)	Co-activation Affinity	34.84 ± 0.000	1.174 ± 0.000	+24.56

**All runs evaluated across Seeds 42, 43, 44 on WikiText-2 (65,536 tokens). External baselines represent zero-shot offline proxies without post-merge fine-tuning.*

Scope Bound & Methodological Safeguards: Evaluation at 48 experts is based on n=3 seeds; paired t-test yields p=0.1330. The advantage over Random-K5 is consistent and directional, but not statistically significant. In exploratory downstream testing (N=100 MMLU), all evaluated merging methods score near the 25% chance floor; capability retention remains unestablished.

OPEN RESEARCH QUESTION FOR NVIDIA DISCUSSION

"Can genuinely pre-merge functional signals identify destructive expert merges without requiring oracle merge measurements?"

I would value feedback on: (1) designing pre-merge capability representations that approximate the low-rank damage space, (2) architectural limits of functional distance across non-uniform routing, and (3) hybrid architectures combining pruning for dormant experts with geometry-guided merging for active clusters.

Traceability: Multi-seed benchmark runs traceable to `reports/BENCHMARK_L40S/`; forensic audit in `reports/FINAL_AUDIT_REPORT.md`.

Target: OLMoE-1B-7B • NVIDIA Technical Exchange